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Dear Hiring Manager,

As a postdoctoral researcher at the University of Trento, I built a real-time audio pattern detector that runs in 14 ms on a Raspberry Pi 4 at F1 0.76, a 74x improvement over the 1038 ms DTW baseline it replaced, published in the Journal of the Audio Engineering Society (2026). That work shipped as a full embedded ML application, not a research demo: streaming audio acquisition, DSP feature engineering, model training and evaluation, and edge deployment, meeting a latency budget comparable in strictness to the constraints an automotive-grade voice-AI product has to hold. That combination of embedded audio inference and hard real-time reliability is what draws me to the Senior Application Developer role on Cerence's industrial voice AI and edge apps team.

My toolkit is built for exactly this kind of work. I use a full DSP feature set (STFT, MFCC, LFCC, PLP, GFCC, S-transform, CQT, onset/beat tracking, YIN pitch estimation) and deploy audio models on embedded platforms through Elk Audio OS, JUCE, VST, and OSC, with C++17 as my primary language for the real-time engines themselves. I also bring a research habit that I expect carries over well to a production edge-apps team: I design frozen-protocol benchmark suites and ablation studies to quantify accuracy, latency, and CPU trade-offs before committing to an architecture, most recently on an edge-to-server offload system (MIRaaS, accepted at IEEE IS2 2026).

I have not worked in the automotive industry specifically, and I want to be upfront about that rather than overstate the overlap; my real-time and embedded-audio background is the closest analog I can point to. On the language side, my French is basic and actively improving (roughly A2), not yet at a working-professional level. I would want to know early whether the role requires daily working French or whether English is sufficient for the technical work, so we can both be clear-eyed about fit.

I am a Canadian permanent resident, eligible to work in Canada without sponsorship, available immediately, and open to being based in Quebec for this role. I would welcome the chance to discuss how my real-time embedded audio ML experience could contribute to Cerence's voice AI and edge applications team.

Sincerely,
Nishal Stanislaus Silva